

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		Addition of water (1) Galactose and glucose correctly drawn (OH below C ¹ on galactose + CH ₂ OH at C ⁶ on glucose) (1)	1	1		2		
		(ii)		Hydrolysis (1)	1			1		
		(iii)		Add Benedict's reagent/solution and heat (1) NOT ref to HCl Change of colour <u>from blue</u> to {yellow/green/orange/brown/(brick) red} (1)	2			2		2
	(b)	(i)		Glucose oxidase has a <u>specific shaped</u> active site (1) Only glucose molecules have a complementary (shape) (1) (Therefore glucose fits) to form an enzyme-substrate complex (1)	1	2		3		
		(ii)		Any two (x1) from: • More glucose the more hydrogen peroxide is produced (1) • Equation shows a one to one relationship / is directly proportional (1) • Concentration of hydrogen peroxide is converted to an electrical signal/ detected by the electrode (1)			2	2		